

Xperience: An Augmented Reality Application to Promote Business to Customer (B2C) Experience in Retail

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Abstract

This research explores the potential of augmented reality (AR) in revolutionizing business-to-customer (B2C) experiences, specifically in the fashion industry within the African retail sector. Customers often struggle to envision how products will look or function in their own spaces due to limited differentiation between products or services. Businesses, on the other hand, face increased competition and struggle to capture customer attention. This makes informed purchasing decisions difficult as customers cannot physically examine products or experience services firsthand. This project aims to create an AR-based shopping experience that is both interesting and convenient, by developing an AR application coined “Xperience”, designed for facial virtual try-on of various fashion items, including glasses, hats, and face masks. With a focus on young, tech-savvy shoppers and businesses looking for a competitive edge, the goal is to build an AR application that enhances the shopping experience and helps businesses boost their sales through AR features such as customizable color options and gesture-based interaction. An agile scrum framework with a participatory design approach was used in the development of the solution. The implementation was done using Unity Engine, 3Ds Max for 3D models optimization and modeling, Figma, Lucidchart and Photoshop for design, and pattern recognition for face landmark detection and occlusion handling. The user testing conducted with 19 participants confirmed that the application is significant and has the potential to improve customer engagement and satisfaction. However, limitations such as small-scale testing, device compatibility issues, and restricted customization options were identified. By combining research, planning, and hands-on development, this work aims to demonstrate that AR can be a game changer for online shopping. The research has significant implications for increased customer loyalty, increased sales, and a better understanding of the strategic value of AR in e-Commerce in a modern business setting.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

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Africhi 2025, Cairo, Egypt
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ACM ISBN 979-8-4007-1849-6/25/11
<https://doi.org/10.1145/3757232.3757236>

Keywords

Business-to-Customer, e-commerce, Augmented Reality, Customer Experience, Product Visualization

ACM Reference Format:

Blessing Song Tang, Omowunmi E. Isafiade, and Kessel Okinga Koumou. 2025. Xperience: An Augmented Reality Application to Promote Business to Customer (B2C) Experience in Retail. In *The 5th Biennial African Human Computer Interaction Conference (Africhi 2025)*, November 04–08, 2025, Cairo, Egypt. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3757232.3757236>

1 INTRODUCTION

Traditional modes of interaction, such as physical storefronts, print catalogues, and face-to-face engagements, have long dominated the business-to-customer (B2C) experience. However, these methods often present limitations, especially regarding time consumption, accessibility, and consumer satisfaction [18]. Modern consumers increasingly seek dynamic and personalized experiences that enable them to make informed decisions quickly and conveniently [24]. The shift to digital solutions was accelerated by the COVID-19 pandemic, which drastically altered consumer behaviour, pushing more interactions into the online realm [16]. However, online shopping also poses significant challenges, especially in the fashion industry, where the inability to try on or interact with products often leads to dissatisfaction due to unmet expectations [38]. Sarka et al [32] pointed out that physical stores can be time-consuming and inconvenient, while online shopping can lead to dissatisfaction due to discrepancies between product images and the actual product when delivered. Immersive technologies, such as AR and Virtual Reality (VR), provide a promising solution to these challenges by creating more engaging, interactive, and informative shopping experiences. These technologies have impacted several industries, including education [20], financial services [26], tourism [13], and healthcare [30], by enhancing visualization and interactive capabilities. While VR immerses users in a fully virtual environment, such as enabling a customer to virtually explore and customize a car showroom to experience different models and features before making a purchase decision, it can be costly to implement and deploy due to the need for equipment such as head-mounted displays (HMDs). On the other hand, AR stands out as a more practical and cost-effective tool for businesses as it overlays digital content in real-time onto the physical world and comes in handy on mobile devices such as smartphones, making it more accessible and functional for everyday use [33]. Furthermore, AR technology has been suggested to enable consumers to virtually try on clothing, accessories, or even makeup in real-time, simulating how products will look on

their bodies or in their personal space before making a purchase. With this personalized and interactive shopping experience, AR can significantly reduce the uncertainty customers often face when shopping online by offering a more immersive and fun shopping experience [29]. Recent advancements in mobile technology have further facilitated the adoption of AR in the retail sector. Today, most smartphones are equipped with cameras and sensors that support AR applications, making it possible for businesses to integrate AR solutions into their mobile apps and websites [29]. Retail giants like Ingvar Kamprad Elmtaryd Agunnaryd (IKEA) and Sephora have adopted AR technologies to allow customers to visualize how furniture would look in their living spaces or how cosmetics would appear on their skin tone, respectively [25]. Yet, despite its immense potential, the widespread adoption of AR in retail has been limited, particularly in small fashion businesses [28], where accurate product visualization, real-time interaction, and user-friendly interfaces remain significant hurdles. Hence, this is the motivation for this research.

1.1 PROBLEM STATEMENT AND PROPOSED SOLUTION

Problem statement:

The fashion sector faces challenges due to the highly personalized nature of clothing and accessories, where factors such as color change are crucial [3]. However, most e-Commerce platforms rely on static images and generic size guides, which fail to capture these nuances and create an immersive shopping experience. As a result, consumers are often reluctant to buy without trying on the items, creating a significant barrier to online shopping. This hesitation can reduce purchase rates, and when purchases are made without proper evaluation, they frequently result in high return rates and negatively impact customer satisfaction [2]. To bridge this gap, AR can be used to allow customers to virtually try on garments and accessories of different kinds to assess how they fit and look in real-time.

In summary, the following are the key challenges.

- Difficulty accurately visualizing how products will fit or appear on the user.
- Static images and size guides do not capture the real-life fit, color, and style nuances.
- Discrepancies between product images online and the actual product can lead to customer dissatisfaction.

1.1.1 Proposed solution: This work proposes a user-friendly and easy-to-use mobile AR application, coined Xperience, designed for facial virtual try-on of various fashion items, including glasses, hats, and face masks. The solution utilizes advanced techniques in facial landmark detection and occlusion management to deliver precise and realistic visualizations. Furthermore, it offers features such as customizable color options and gesture-based interaction.

1.2 AIM AND OBJECTIVES

This work aimed to develop and evaluate the Xperience facial virtual try-on AR application for promoting business-to-customer (B2C) experience. To achieve this aim, the following are the objectives of this study:

- To develop the Xperience application to address key technical and user experience challenges in the fashion industry.
- To evaluate the effectiveness of the developed application through various system-level testing.

1.3 SIGNIFICANCE OF THE STUDY WITHIN THE AFRICAN RETAIL MARKET

This study addresses key challenges in Africa's emerging small e-Commerce and fashion retail sectors by presenting a functional, user-friendly mobile solution named Xperience, serving as a proof of concept. The application was developed to enable users to virtually try on fashion items such as glasses, hats, and masks. The Xperience application offers realistic visualization that helps users assess how products suit their skin tone, face shape, and style preferences [4].

The following are the potential benefits and long-term implications of this study:

- Allow customers to virtually try on products before purchase.
- Enhance product visualization and interactivity through AR.
- Offer customizable colour options and gesture-based interaction to enhance the user experience.
- Demonstrates the potential for broader adoption of AR across the fashion industry.
- Serves as a scalable solution that could be adapted across other sectors within African retail, such as cosmetics.

2 RELATED STUDIES

2.1 AR AND VR IN RETAIL

In today's digital landscape, customer experience (CX) has become a key value for businesses as immersive technologies, such as AR and VR, rapidly transform the way businesses interact with their customers [5]. Immersive technologies extend or create a new reality by leveraging 360 environments [34]. In this context, this section reviews research on AR and VR in e-Commerce, focusing on their ability to enhance customer engagement, and improve product visualization.

Chen et al. [6] conducted a comprehensive literature review investigating the impact of AR and VR on e-Commerce customer behaviour. Their analysis encompassed numerous studies exploring the engagement and practical applications of these technologies within online shopping contexts. The review concluded that these technologies significantly enhance customer engagement compared to traditional online shopping methods. By enabling customers to experience products in a more realistic and immersive manner, such as virtually trying on clothes or visualizing furniture in their homes, the technologies effectively address key e-commerce challenges, including high product return rates, by fostering increased customer confidence and satisfaction. Similarly, Khaldy et al. [15] evaluated re-branding e-Commerce using AR and VR. They investigated the latest AR and VR to see what it means for online shopping. The study found that these technologies give high levels of interactivity and immersion. As a result, consumers can experience products in a virtual space as if they were real. In addition, Jayawardena et al. [31] investigated AR and VR in marketing specifically through video ads. Their research showed that immersive technologies ads

increased consumer engagement and interest in buying products because of the realistic presentation of products. While insightful, their work primarily examined the impact of AR and VR in advertising, leaving a gap in understanding their broader application across the entire B2C journey.

Several studies [5, 6, 15, 31, 34] have shown that AR and VR technologies enhance the shopping experience, increase sales, reduce marketing costs, improve customer satisfaction and loyalty, which is good for business. Singh et al. [36] also investigated AR and VR for interactive and immersive shopping experiences. Their research provided numerous examples of how these technologies are transforming online retail into a more efficient, personalized experience.

2.2 AR ON CONSUMER ENGAGEMENT AND BUSINESS OUTCOMES IN E-COMMERCE AND RETAIL

Khan [17] has evolved from a new concept to a must-have tool in e-Commerce, improving customer satisfaction and operational efficiency. A study by Jawale et al. [4] asserts that AR reduces product return rates by allowing consumers to see the product in real time before purchase. According to Hoffmann and Mai [11], AR enhances the shopping experience of customers by providing immersive interactions. Park and Kim [27] studied the effect of AR on purchase intentions by looking into cognitive elaboration and shopping goals. They discovered that the technology increased sales by allowing consumers to make faster and more confident purchase decisions. Interactivity and cognitive engagement were the key drivers of higher sales. Nevertheless, their study concentrated on the pre-purchase phase, offering limited insights into post-purchase engagement and long-term customer relationships facilitated by AR. Wedel et al. [39] pointed out that AR technologies change consumer behaviour by introducing new ways to interact with products. Their research supports the consensus that AR can improve consumer engagement with products and ultimately increase sales and profits for businesses. However, the study provides a broad overview without delving into specific AR application designs as done in the current research, and their differential impact on various stages of the B2C experience.

2.3 GAPS IN LITERATURE REVIEW

Although there has been a lot of research on AR in e-Commerce and retail, there are still some gaps. Chen et al. [6] noted a lot of short-term effects of AR and VR, such as increased interactivity and engagement with products, but long-term effects on loyalty and business are yet to be identified. Furthermore, their review did not specifically focus on the holistic B2C experience, encompassing aspects beyond the immediate transaction. Similarly, Khaldy et al. [15] agree with Chen et al. [6], noting a research gap in AR application across e-commerce sectors. These researchers pointed out that while much of the research has focused on retail, there is a need for further investigation into customer service and personalized marketing. Their work highlighted the potential but did not delve into the design principles and frameworks for AR applications that can effectively integrate various touchpoints of the B2C journey [40]. Furthermore, Hoffmann and Mai [11] proposed that more research

be done on the privacy of AR and its access to user data. Hence, there is a need for additional research on AR's impact on user data privacy and security. Moreover, their future research directions did not explicitly address the design and evaluation of AR applications aimed at fostering a continuous and enriched B2C experience.

In summary, these studies collectively highlight the profound impact of AR technologies across various domains, emphasizing their potential to revolutionize consumer experiences and drive business success through enhanced engagement and interactivity. However, a significant gap remains in the explicit investigation of how AR applications can be strategically designed and implemented to enhance the B2C experience, moving beyond singular touchpoints like initial purchase or advertising. Therefore, the present study seeks to address these gaps by developing and evaluating Xperience, a virtual try-on AR application designed to overcome key technical and user experience challenges in fashion e-commerce, and by assessing its effectiveness through various system-level testing.

3 DESIGN AND METHODOLOGY

3.1 THEORETICAL FRAMEWORK

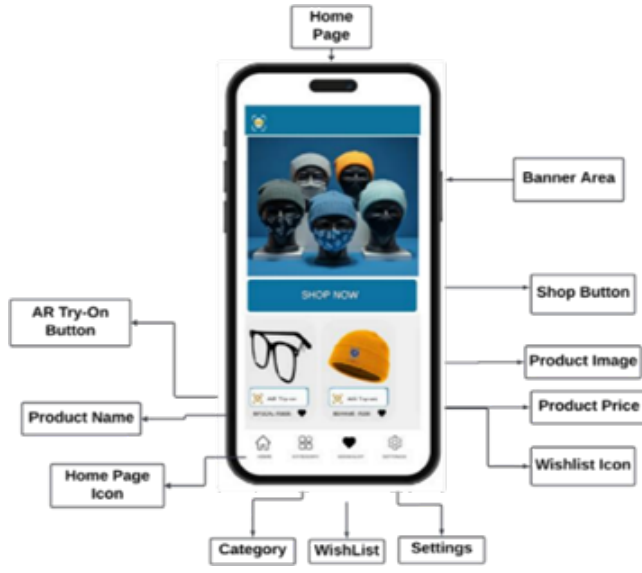
This research used a participatory design approach alongside throw-away prototyping and the Agile Scrum framework to develop the Xperience mobile application. Participatory design was adopted to allow Scrum teams to be involved throughout the prototype design process [22]. Scrum teams consist of the scrum masters, the main supervisor of this research, and the product owner. The development tasks were conducted by the development team, which was responsible for designing, developing (coding), testing, and delivering the solution. To support early-stage ideation, a throw-away prototyping method was employed for designing the initial version of the application [12] (see Figure 1(a)). Furthermore, Figure 1(b) presents the navigational flowchart of the proposed system. The decision to use an agile approach, and specifically the Scrum framework (Figure 2), was driven by its adaptability and focus on incremental delivery [1][10]. Scrum helped the team maintain clear communication and make steady progress through short development cycles. As each incremental version builds upon the previous one, at the start of each sprint, the team carefully reviews and prioritizes items in the product backlog, including user stories and bugs found in earlier releases, by considering the outstanding bugs found after each iteration. Feedback from the team at the end of each sprint played a key role in shaping future improvements.

3.2 CONCEPTUAL FRAMEWORK

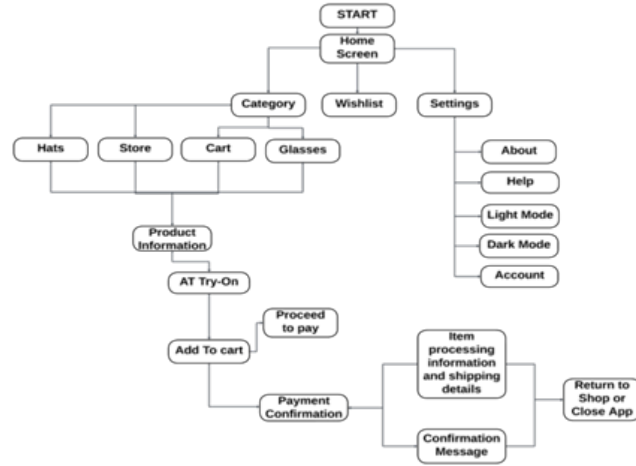
The overall framework of the AR try-on solution is presented in Figure 3. This framework encompasses: (a) Data collection and pre-processing (Bottom layer); (b) Virtual try-on and AR system (Middle layer); and (c) End user engagement (Topmost layer).

3.2.1 DATA COLLECTION AND PRE-PROCESSING (BOTTOM LAYER).

The data collection entails relevant 3D models, which were obtained from platforms such as TurboSquid and Sketchfab. Models were optimized using the 3Ds Max to ensure that they are compatible with Unity's requirements. This optimisation was done by reducing the number of vertices (individual points in 3D space that define



(a) Prototype of the homepage of the application



(b) Navigational flowchart

Figure 1: User Interface (UI) Prototype and Navigational Structure of the Application

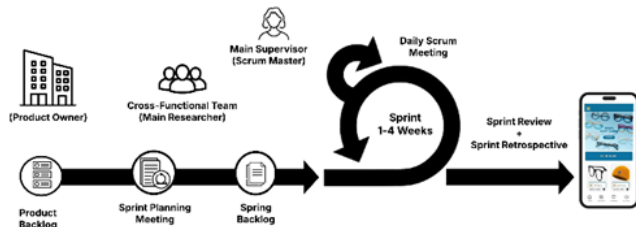


Figure 2: Scrum Framework for Xperience Application

the shape of a 3D model) and polygons (flat surfaces that connect the edges of a 3D model) [22].

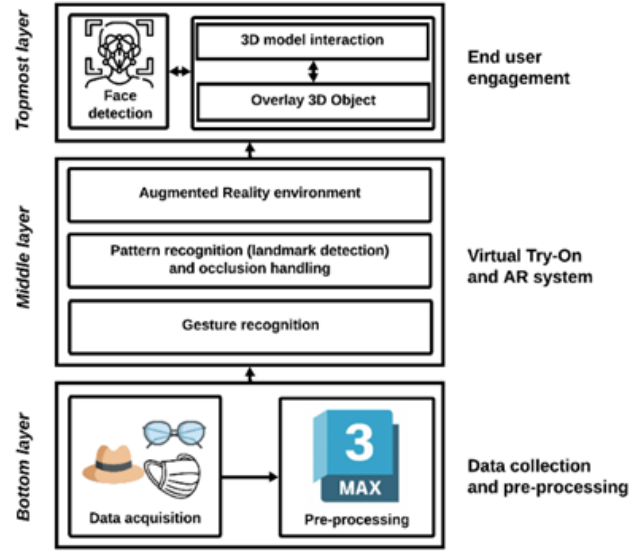


Figure 3: Overall system framework of Xperience

3.2.2 VIRTUAL TRY-ON AND AR SYSTEM (MIDDLE LAYER).

- (A) **AR Environment:** This was set up within Unity. Key aspects include [19] [8]:
- (i) **Scene setup:** Defines the virtual objects, their initial positions, and their interactions with the real world.
 - (ii) **ARCore configuration:** We used ARCore in our Unity project by configuring settings such as camera parameters. ARCore was essential in enabling the AR experience, as it supported the implementation of key functionalities including motion tracking, and light estimation. Additionally, the ARCore SDK allowed us to access device sensors, process camera frames, and render virtual content within the real-world environment.
 - (iii) **Interaction logic:** This involves Unity scripts that manage user interactions, including: a) color change; and b) face movement.
 - a) *Color change:* A list of colors was predefined to allow the user to select whichever color they want at random.
 - b) *Face movement:* This functionality is enabled through real-time face tracking (face landmark detection), which utilizes embedded facial landmark detection. Accordingly, this feature allows users to move their face freely while the virtual asset remains aligned.

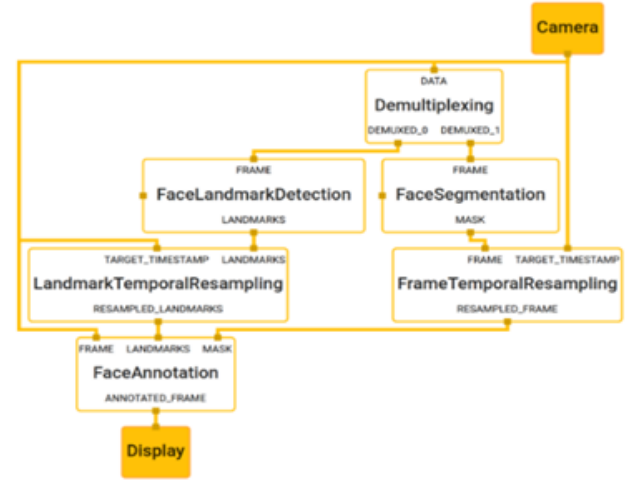
(B) Pattern Recognition (Landmarks) and Occlusion

- (i) **Face landmark detection:** To enhance the implementation of the AR try-on feature, AR Core was used because of its Convolutional Neural Networks (CNNs) model and MediaPipe library, including the Face Mesh for face detection [41]. In our solution, the MediaPipe's Face Mesh identifies the face in real-time, locating specific landmarks, such as the corners of the eyes, the tip of the nose, and the edges of the lips [7]. Figure 4(a) presents a MediaPipe flowchart

that performs face landmark detection with portrait segmentation (isolating the face from the rest of the scene). Figure 4(b) presents the generated 3D mesh based on the identified landmarks. This mesh provides a foundation for overlaying AR content, such as glasses and hats. In our application, a predefined face mesh asset called AR Face Filter was used to create a realistic representation of the face [34]. Moreover, the ARFaceManager component in ARcore was used, as it works by creating GameObjects for each face detected in the AR environment. The system allows only one face ($n=1$) to be tracked simultaneously. This component also allows the placement of 3D objects like glasses onto a user's face. Figure 5(a) illustrates how different objects are placed using the ARFaceManager, showing how glasses and hats are applied at various angles. The prefab approach facilitated the placement of 3D objects on the face. The following elements were fundamental to this process:

- **Face Prefab:** This was used to dynamically adjust the prefab.
 - **AR Default Face:** Serves as a predefined face model, providing the foundation for face tracking.
- (ii) **Occlusion handling:** Occlusion describes the process of preventing the rendering of objects that are completely hidden from the camera's view by other objects. This technique is commonly referred to as "Occlusion Handling". To effectively manage occlusion, assets often employ techniques like adjusting the transparency of overlapping elements (see Figure 5(b)), re-evaluating the visibility of the user's face and overlaying the AR object on the face to complete the process. For example, the transparency of glasses might be adjusted where they overlap with the user's face, ensuring the mesh does not appear to penetrate the user's skin [23]. The Flowchart in Figure 6 outlines a process for handling the visibility and positioning of the AR object to a user's face. Once the process begins, the user's face is detected, and the system checks if the user's face is fully visible. If it is not, the process moves to the next step, where the process skips to overlay the AR object, after which, if there is any obstruction detected, the process moves to adjust transparency; if not, the process moves to adjust the position of the object.
- (iii) **Gesture recognition:** This allows users to interact with a 3D model in an AR try-on feature. This is achieved through the integration of computer vision, ML, and AR frameworks, enabling real-time detection and interpretation of hand gestures and facial movements [35]. Camera input is pre-processed, including noise reduction and normalization, to ensure compatibility with detection algorithms. Facial movements, such as head nods and tilts, are analyzed through facial landmark mapping and used to manipulate the 3D model, enabling actions like aligning it with the user's face [7]. The AR framework anchors the 3D model to the user's facial features and continuously adjusts it based on tracked gestures for seamless integration [9].

3.2.3 END USER ENGAGEMENT. This final layer represents the activities of the individuals who will be using the application. At this stage, it is crucial to emphasize that the final product will be



(a) Face landmark detection and segmentation using MediaPipe [7]



(b) Landmark detection and segmentation portrait [7]

Figure 4: Landmark Detection and Segmentation of Facial Portraits Using MediaPipe

in the form of an APK extension, which users can easily install on their Android devices. User interaction with the Xperience mobile AR application involves engaging in the facial virtual try-on experiences for fashion accessories. The application superimposes digital representations of fashion items, such as eyewear, hats, and face masks, onto the user's face or head. This real-time overlay enables users to visualize products as they would appear in reality, offering a more realistic and immersive shopping experience. Such interactivity not only enhances product engagement but also reduces the uncertainty typically associated with traditional 2D, non-interactive online fashion platforms.

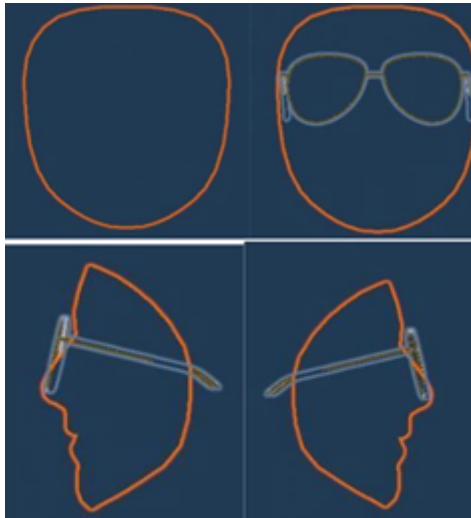
3.3 APPLICATION EVALUATION

To ensure that the Xperience application worked as expected, the following tests were conducted [12]:

A. DEVELOPMENT TESTING



(a) Example of 3D object placement on a user's face via ARFaceManager



(b) Occlusion: transparency adjustments to prevent the mesh from showing where glasses overlap with the user's face

Figure 5: Transparency Adjustments and 3D Facial Overlay via ARFaceManager

- (i) **Unit testing:** This testing was considered to ensure that the methods or functions of the classes implemented are operating as intended.
- (ii) **Component testing:** This testing focused on verifying the functionality of the integrated units that form a component, to ensure they are operating as intended. In this case, we focused the testing on the component interfaces to prevent issues such as parameters being passed in the wrong order.

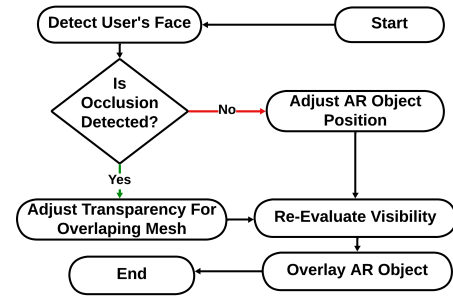


Figure 6: Occlusion Handling Process

- (iii) **System testing:** All components are integrated and tested as a complete program to ensure that the integrated components are compatible and interact correctly. During system testing, the emphasis was on verifying component compatibility, correct interaction, and timely, accurate data exchange through system interfaces.

3.4 PARTICIPANT SELECTION AND DEMOGRAPHICS

The participants recruited for this study were individuals aged above 18 years (consenting age), with a primary focus on tech-savvy users. This age group was selected based on their familiarity with digital technology. All participants voluntarily consented to take part in the study and were informed of the study's purpose, procedures, and data handling. The evaluation criteria used to measure user experience and satisfaction included:

- **Ease of navigation:** Evaluated by the time participants took to complete key tasks within the application, such as navigating through the interface and completing virtual try-ons.
- **AR object placement accuracy:** Feedback was gathered from participants regarding the accuracy of virtual item placement, such as the alignment of virtual fashion items on the face and head.

Based on the above, User satisfaction and overall experience were captured using a five-point Likert scale, which included questions on usability and the likelihood of recommending the application to peers. Table 1 presents the key questions used to evaluate these factors across the dimensions of Perceived Ease of Use, Behavioural Intention of Use, and Perceived Usefulness.

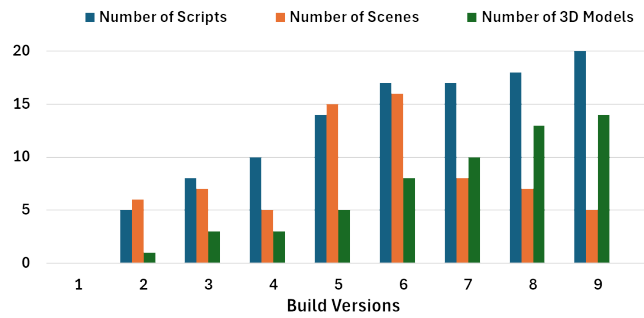
4 RESULTS AND DISCUSSION

This work developed a mobile AR application called Xperience for virtual try-on to provide users with an immersive and engaging shopping experience. The application was developed on the Windows 10 Pro operating system, running on an Intel(R) Core(TM) i7-6700K CPU at 4.00 GHz with 32 GB of RAM. The Unity game engine (version 2022.3.21f1) and 3Ds Max 2023 were used, and C# scripts were implemented using Visual Studio 2019. The application runs on Android devices (version 10 and above). Figure 7 shows the cumulative increase in the number of scripts, scenes and 3D multimedia data used during the development of this application as

Table 1: Key Evaluation Questions

Aspect evaluated	Key Question
Perceived Ease of Use	How keen are you on exploring other fashion items available in the Xperience app beyond what you tested? (Not Keen, Somewhat Keen, Keen, Quite Keen, Very Keen)
Behavioural Intention of Use	How likely are you to recommend the Xperience app to your friends or family? (Never, Unlikely, Likely, Most Likely, Extremely Likely)
Perceived Usefulness	How satisfied are you with the overall features in the Xperience application? (Not Satisfied, Somewhat Satisfied, Satisfied, Quite Satisfied, Extremely Satisfied)

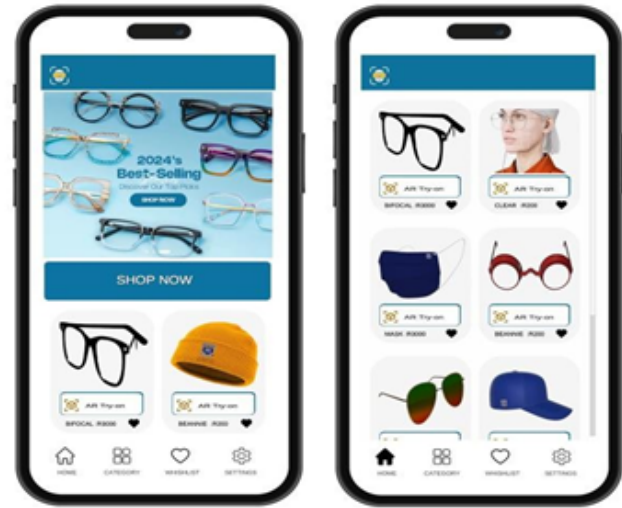
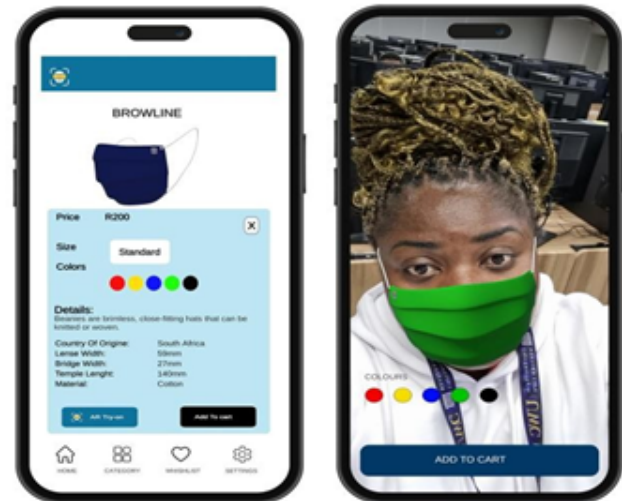
different builds of the Android Package Kit (APK) emerged before the final product.

**Figure 7: Total Number of Scripts, Scenes, and 3D Models Utilised**

This research successfully developed a user-friendly and easy-to-use mobile AR application to enhance online shopping experiences through virtual try-ons of eyewear, hats, and face masks. The application aims to reduce uncertainty in online shopping by providing users with an immersive and interactive experience. The Xperience application includes key features such as virtual try-ons, colour customization, and intuitive navigation.

Xperience navigation icons guide users through different sections of the application, including Home, Category, Wishlist, and Settings. The application's homepage provides easy access to the store using a "Shop Now" button, while the trial page allows users to interact with selected items in real time.

Tapping the "Shop Now" button directs the user to the Shop page. The "AR Try-On" button, shown in Figure 8(a), navigates the user to the Try-on page, where they can virtually try on the selected item. The application then superimposes the virtual item directly onto the user's face or head (see Figure 8(b)). In comparison to the AR-based virtual try-on by Liu et al. [21], where users engage in virtual try-on by detecting the ground plane and placing a life-sized virtual body into a real-life scene, a similar approach is adopted by Hwangbo et al. [14], however, the Xperience solution takes a different approach. According to Liu et al. [21] and Hwangbo et al. [14], their method allows users to view a personalized, life-sized virtual body with garment models, augmented with animations such as posing or

**(a) Home and store pages****(b) AR try-on with mask try-on feature (application of face detection)****Figure 8: Home and Store Navigation with AR Try-On and Face Mask Detection**

walking in the real world (see Figure 10). Moreover, in the Xperience solution, the try-on page offers color options for each item, enabling users to choose a shade that complements their skin tone before making a purchase (as shown in Figure 8(b)). In contrast, Liu et al.'s solution allows users to select various clothing styles, which are displayed statically. According to the authors, this enables users to match garments on their personalized virtual avatar with predefined animations. The Xperience application has a navigation bar with icon options to guide the user to different pages in the store: (i) Home, (ii) Category, (iii) Wishlist, and (iv) Settings, as shown in Figure 9. In comparison, the solution proposed by Liu et al. [21] does not report additional features related to navigation or similar user interface elements.

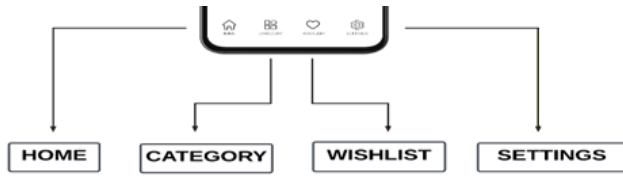


Figure 9: Xperience navigation icons



Figure 10: AR-based try-on [21]

4.1 APPLICATION TESTING

The tests are intended to examine whether the application is doing what it is intended to do and to discover any potential defects (bugs) in the program before it goes into operation. Therefore, the Xperience application went through two stages of testing, such as development and user testing [39]. Specific performance metrics were used to evaluate test outcomes, including response time for navigation is approximately 2 seconds, and functionality success rate for buttons and features was satisfactory. These metrics ensured the application met usability and reliability standards. During testing, unexpected outcomes were observed, such as the mask material appearing incorrectly during AR try-on due to occlusion issues and navigation buttons failing to function in AR mode. These outcomes prompted design iterations, including modifying the material to handle occlusion effectively and updating button functionality to ensure seamless navigation. In what follows, further details on the tests conducted are provided.

4.1.1 DEVELOPMENT TESTING.

- (i) Unit testing: All sections of the try-on application that require user input were tested. For instance, Table 2 presents a short list of test cases considered during the development phase for unit testing on the account creation. We used the inputs that could enable the registration section to generate error messages resulting from incorrect information entered. Finally, we compiled a list of data that could cause the registration input field buffers to overflow [37].
- (ii) Component testing: As the application accesses the functionality of the objects through interfaces, we implemented various types of interface testing to ensure reliable interaction between components. Interface testing is essential for detecting potential issues such as incorrect usage, mismatched expectations between components. By thoroughly testing these interfaces, we aimed to identify problems like

Table 2: Unit Testing on Account Creation

Iteration	Input	Expected Results	Bug Fixed
1	Azamah 12345678	Arsenal12345678 was ignored by the input field	Only letters (alphabets) can be entered
2	Afro Short Azamah	Inputs are longer	Updated to only take inputs that fit in the field
3	Register Button	No display	Updated to display a popup

passing an incorrect data type and incorrect parameter ordering [12]. Table 3 presents some test cases designed as part of component testing.

Table 3: Home Component Testing

Iteration	Input	Expected Results	Bug Fixed
1	Verify if the slider areas work correctly	The slider moved both ways	Modified to slide left and right
2	Verify that navigation went to the correct pages	Did not return to the previous page	Modified to return to the previous page

- (iii) System testing: All components developed were integrated to create a final application version. Only the interactions between the components and objects that make up the system were considered during the system testing. The reasons behind the system testing are to check that the components, such as Home, Category, Wishlist, Settings, and AR, are compatible, interact correctly, and transfer the right data at the right time across their interfaces [12]. Table 4 presents some test cases designed as part of system testing.

4.1.2 USER EXPERIENCE EVALUATION. To assess the application's effectiveness, we conducted a small-scale user acceptance test with 19 participants after due consent was obtained, which yielded a positive response. The participants in this test were young adults of consenting age, between 21 and 35 years old, with 8 females

Table 4: AR System Testing

Iteration	Input	Expected Results	Bug Fixed
1	Verify if the camera opens when AR mode is active	The camera opened with an AR	None
2	Verify that the mask is shown on the face during try-on	The mask showed when the mesh material was white	Modified the material to handle occlusion to avoid the mask from showing

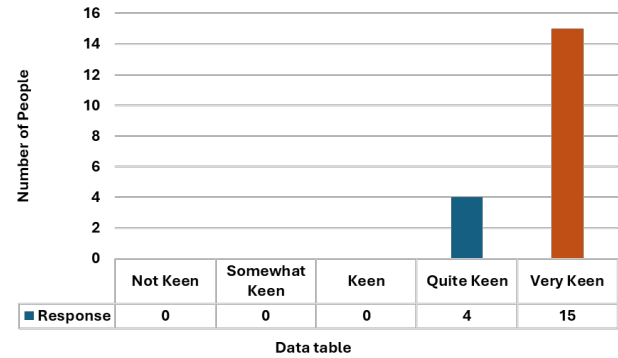
and 11 males. Figure 11(a) shows 75% (15) of respondents are “Very Keen” and 20% (4) are “Quite Keen” to explore additional fashion items in the Xperience application. No one selected the low-interest options. Overall, 95% expressed high enthusiasm to explore more fashion offerings beyond what they tested. Figure 11(b) presents, 55% (11) of respondents would “Definitely” recommend the app, while 40% (8) are “Most Likely” to recommend it. Only 5% (1) selected “Likely”, and no one chose the negative options. Overall, 95% of users expressed a strong likelihood (either “Most Likely” or “Definitely”) to recommend the Xperience app to friends or family. These robust findings demonstrate the Xperience application’s potential to disrupt traditional fashion browsing behaviors and establish a new paradigm for digital engagement.

4.2 EVALUATION OF RESULTS

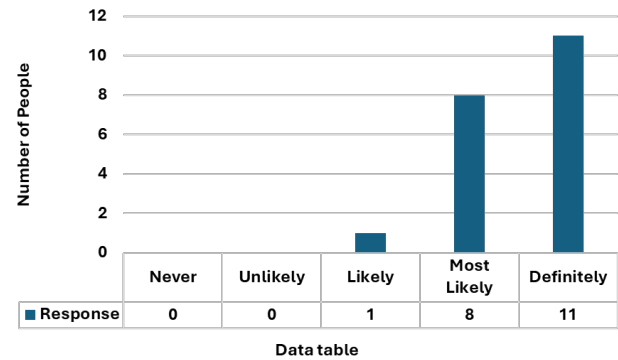
This study aimed to develop and evaluate the Xperience application. Features such as virtual try-on, and colour selection tools to support personalized shopping fulfilled the first objective, which is to develop the Xperience application to address key technical and user experience challenges in the fashion industry. Objective 2 (To evaluate the effectiveness of the developed application through various system-level testing), centered on evaluation, was addressed through development, user and system testing. Test results confirmed that the application functions as intended, though some issues (e.g., occlusion errors, navigation failures) were initially identified. These were corrected through iterative development. Compared to other systems like that of Liu et al. [21], Xperience provides a more intuitive interface, though with limited body-tracking capabilities. These findings underscore both the strengths and current limitations of the solution.

5 CONCLUSION AND FUTURE RECOMMENDATIONS

This work developed a successful user-friendly and easy-to-use AR application (coined Xperience) to transform online shopping involving eyewear, hats, and masks, among others, particularly for, but not limited to, Small, Medium, and Micro Enterprises (SMMEs). By allowing customers to virtually try-on products, customize colours, and visualize how they look, Xperience aims to bridge the gap between online and in-store shopping experiences. Traditional online shopping often leads to uncertainty and dissatisfaction due to the inability to physically try on products. Xperience addresses this challenge by developing an immersive and interactive AR mobile application that offers a more engaging and personalized shopping experience, ultimately leading to higher customer satisfaction and increased sales. The feedback from user experience testing confirmed the potential efficacy of the Xperience application with areas for improvement. Potential challenges include the high cost of realistic 3D models and limited user familiarity with AR technology. Strategies to overcome these challenges include leveraging open-source 3D modelling tools to reduce costs and implementing user education initiatives, such as tutorials and onboarding guides, to enhance user adoption and confidence in using the application. The Xperience virtual try-on application is currently a proof of concept with a need for future research and expansion of the application’s functionality to include a wider range of features and



(a) Response from user acceptance test (How keen are you on exploring other fashion items available in the Xperience app beyond what you tested?)



(b) Response from user acceptance test (Would you recommend or use the application?)

Figure 11: User Feedback on App Exploration and Recommendation Intent

products. Future developments of Xperience will focus on expanding its product range, improving the accuracy of virtual try-ons, and optimizing the user experience across various devices. Some future recommendations for this research are:

- (1) *Expanding product categories for scalability:* Future research could focus on expanding the application’s functionality to include a wider range of product categories, such as accessories and footwear.
- (2) *Comprehensive user testing for validation:* To validate the application’s effectiveness as a try-on tool, comprehensive user testing should be conducted. Gathering feedback from target users at a large scale will help identify areas for improvement and assess the application’s impact on purchasing decisions.
- (3) *Data privacy:* Since the application is to collect user data, it will need to implement robust data privacy measures such as regular security audits and compliance with relevant data protection laws to protect user information.
- (4) *Real-time lighting and shadows:* Implementing real-time lighting and shadow calculations would enhance the realism of the virtual items.

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